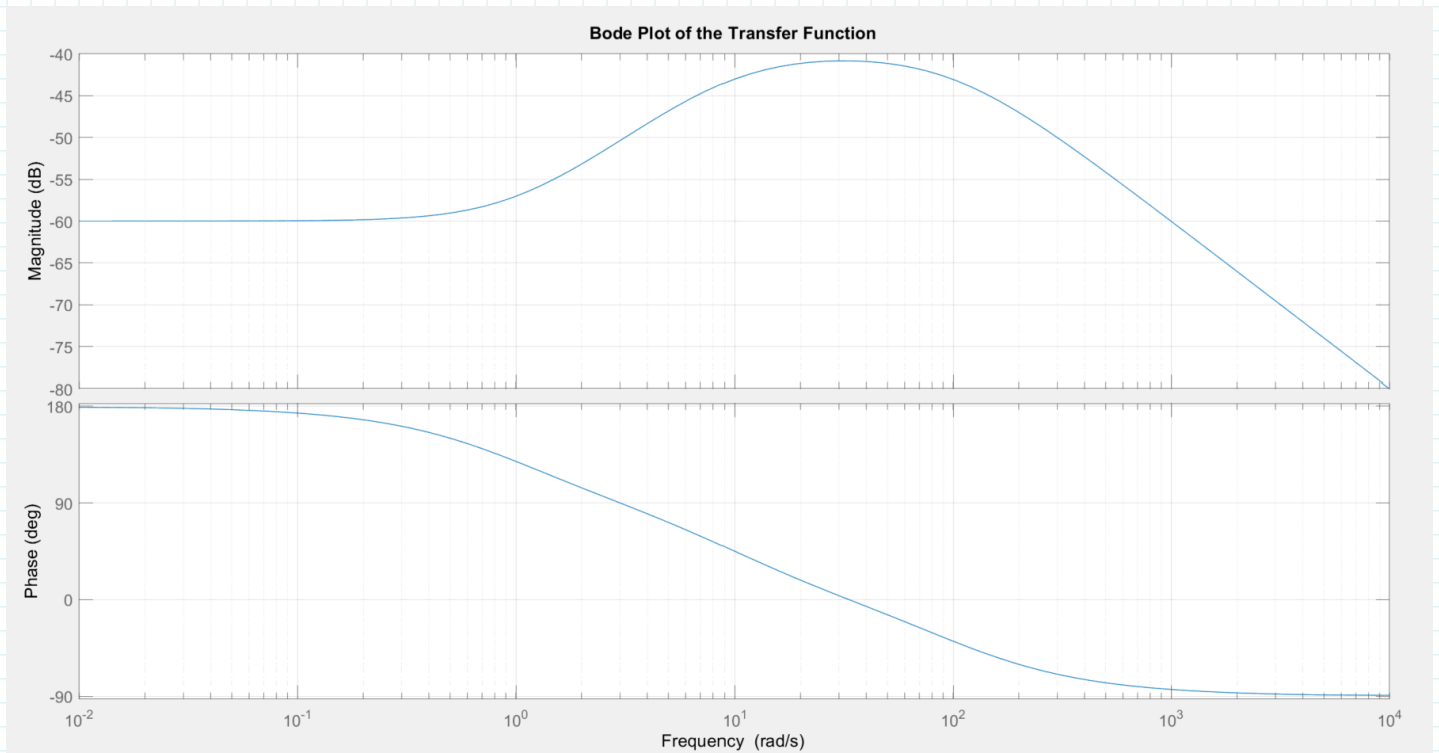
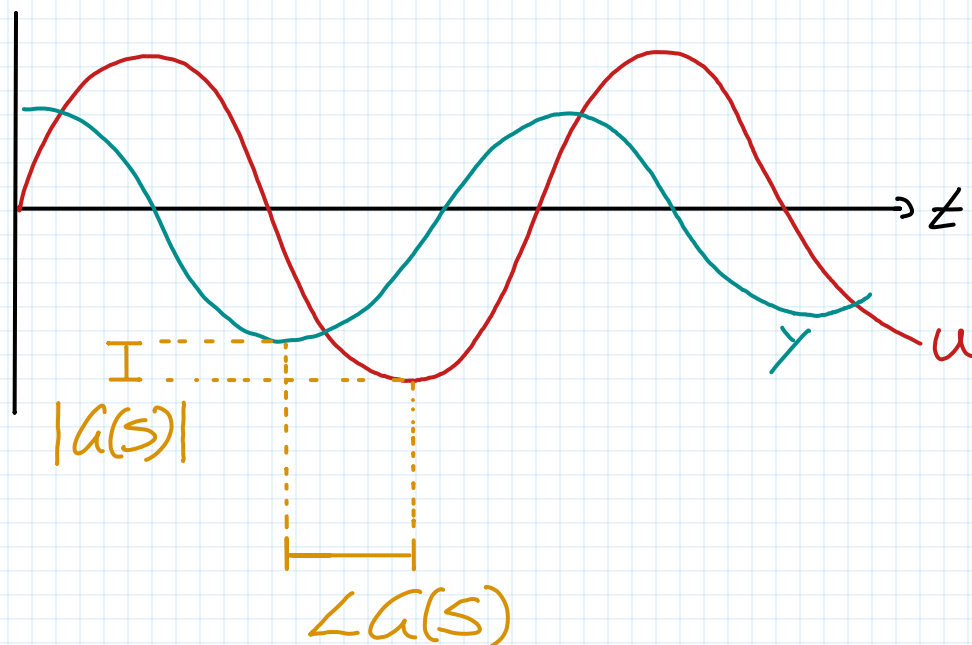


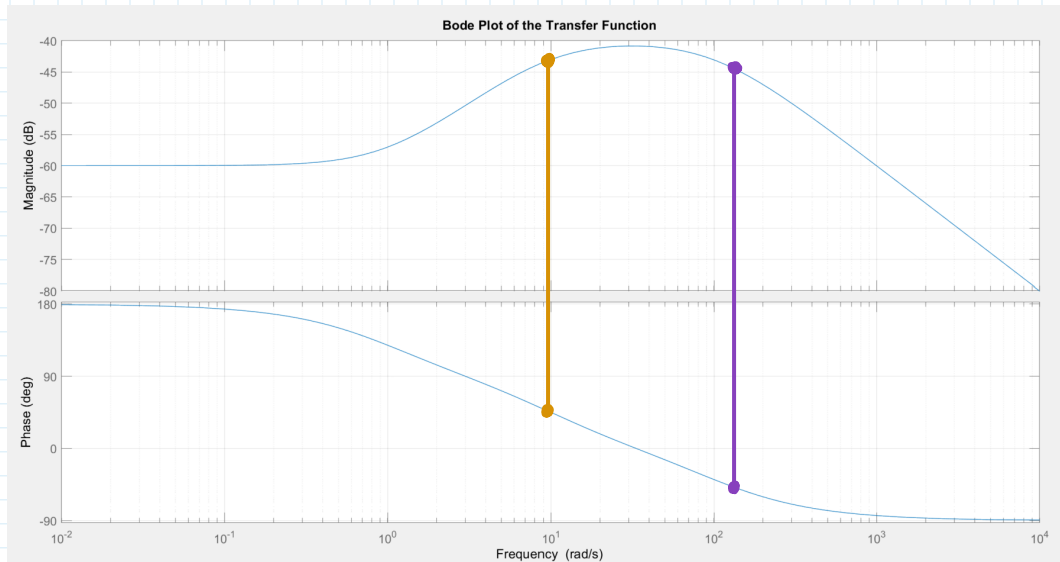
• Last time we looked at the Bode plot.



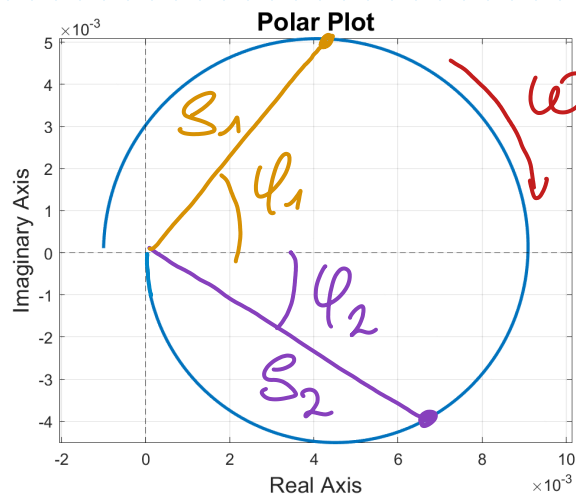
A set of graphs that show magnitude and phase of a system, depending on the frequency input.



Then we saw, that we can summarize both plots in one, by making the frequency implicit.



Make  $\omega$  implicit



This is where we left off. The Polar plot can now be extended to become really useful for stability analysis!

Why do we need something other than Bode plots?

+ Intuitive display of frequency response

- Misleading if

- Open-loop unstable

- Non-minimumphase zeros

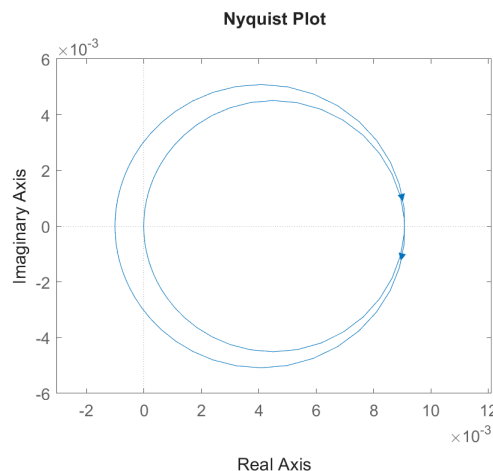
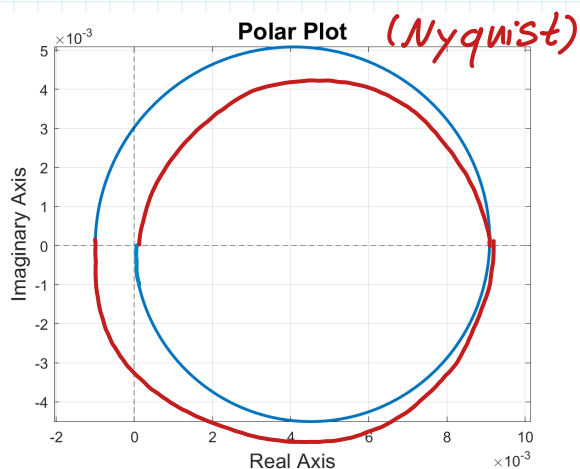
} Nyquist plot is generally applicable  
 $\Rightarrow$  Solves these!

## Nyquist Plot

• The polar plot goes from  $\omega \in [0, \infty]$ .

$\Rightarrow$  The Nyquist plot extends this to  $\omega \in [-\infty, \infty]$ .

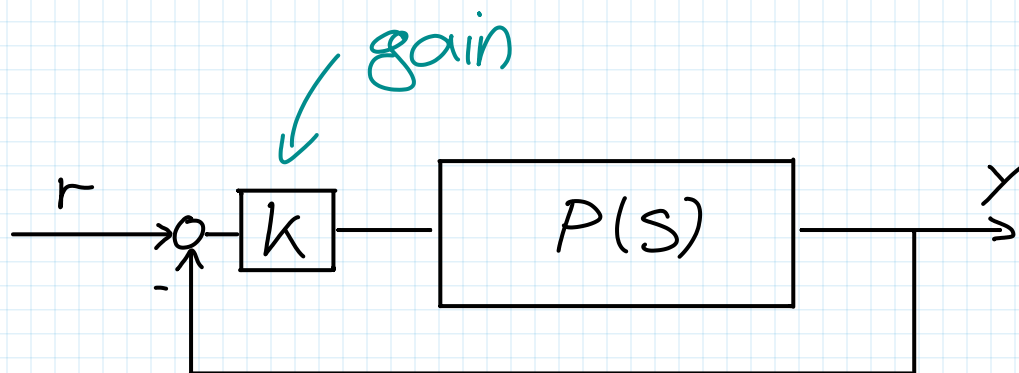
$\Rightarrow$  Graphically this is equivalent to mirroring the polar plot about the real axis.



$\Rightarrow$  Not a lot of new information right?

The Nyquist plot will help us assess closed-loop stability based on the open-loop

Recall standard feedback architecture:



The closed loop has tf  $T(s)$

$$T(s) = \frac{kP(s)}{1 + kP(s)}$$

The closed loop is **unstable**, if  $T(s) = \infty$ .

$$T(s) = \infty \text{ if } 1 + kP(s) = 0$$

$\Rightarrow$  System blows up, if  $P(s) = -\frac{1}{k}$

The  **$-\frac{1}{k}$**  point will be very important!

Note how we are making conclusions about closed-loop stability, based on the open loop!



To determine stability of a closed loop system, we look at the:

## Nyquist Criterion

Given a Nyquist contour  $\Gamma_s$ , let  $P$  be the number of poles of  $G(s)$  encircled by  $\Gamma_s$ , and  $Z$  be the number of zeros of  $1 + G(s)$  encircled by  $\Gamma_s$ . Alternatively, and more importantly, if  $Z$  is the number of poles of the closed loop system in the right half plane, and  $P$  is the number of poles of the open-loop transfer function  $G(s)$  in the right half plane, the resultant contour in the  $G(s)$ -plane,  $\Gamma_{G(s)}$  shall encircle (clockwise) the point  $(-1 + j0)$   $N$  times such that  $N = Z - P$ .

~ Wikipedia

• Largest point of confusion in CSI...

Let's break it down:

• This equation always holds:  $N = Z - P$

$Z$ : # of unstable closed-loop poles

$P$ : # of unstable open-loop poles

$N$ : # of clockwise Nyquist-plot encirclements of the  $-1/k$  point.

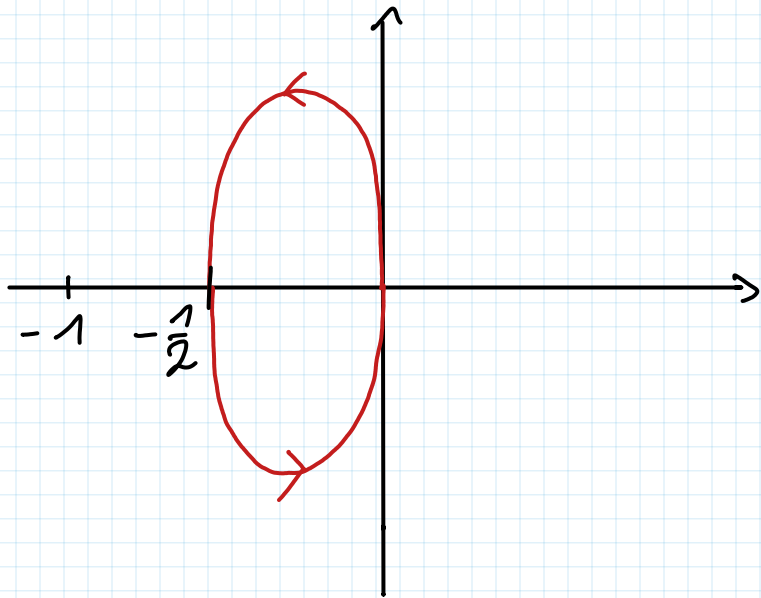
$\Rightarrow$  For a stable closed loop, we require  $Z = 0$ .

Therefore we need  $N = -P$

Ex:

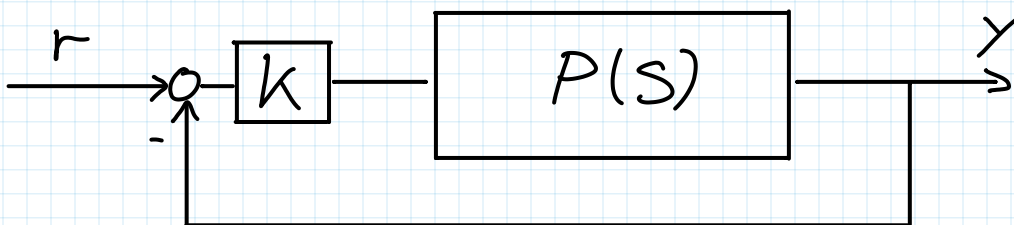
Given  $P(s) = \frac{1}{s-2}$  } unstable pole!  
 $\Rightarrow P = 1$

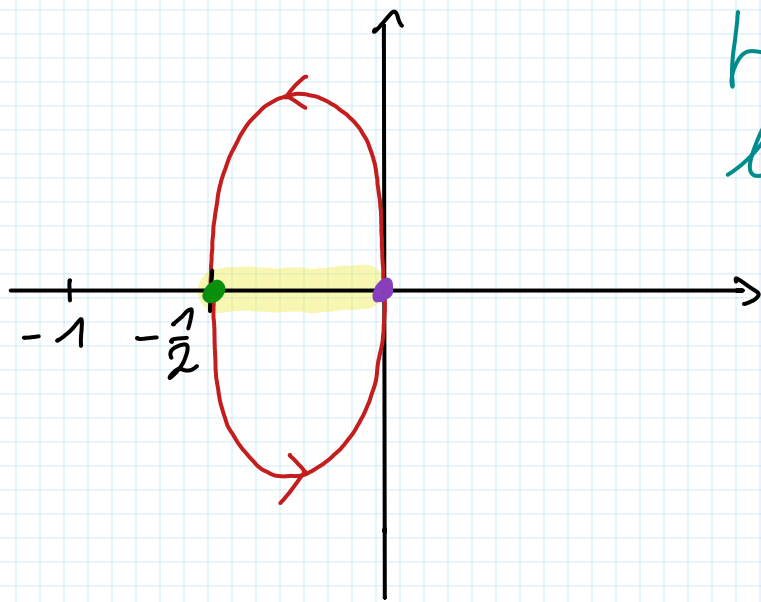
Plotting Nyquist of  $P(s)$  we get:



For stability we  
 require  $N = -P = -1$

We need Nyquist to encircle  $-\frac{1}{K}$   
 $-1$  times (or once in counter-  
 clockwise direction!)





Here:  $-\frac{1}{k}$  must lie in **this** realm

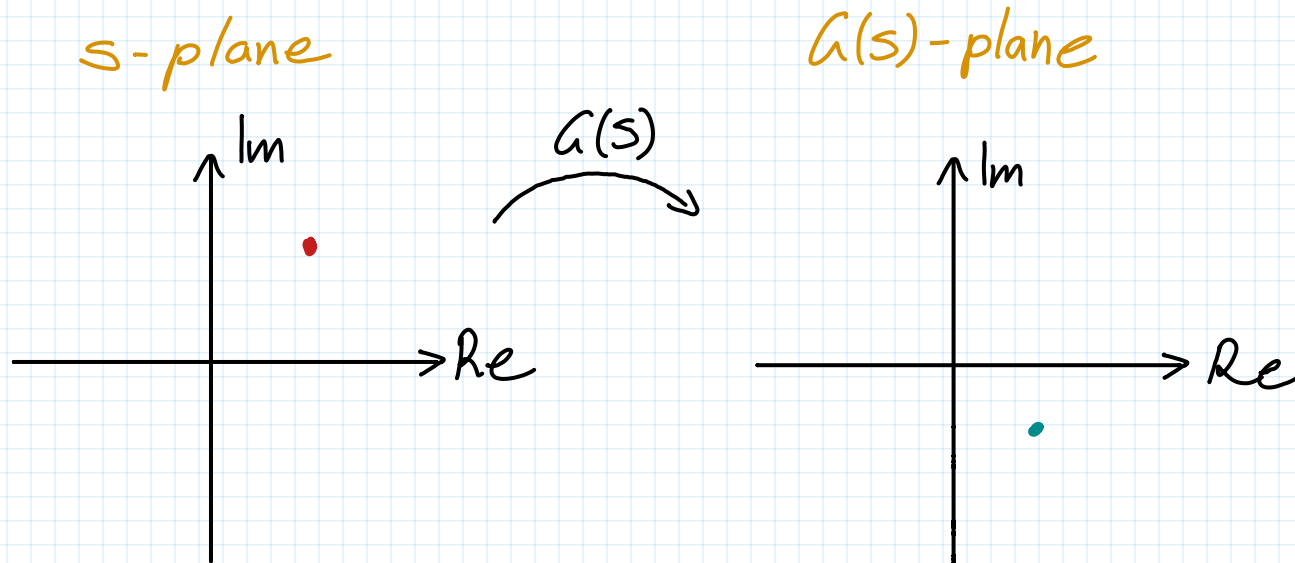
Condition for stable closed loop:

- $-\frac{1}{k} < 0 \Rightarrow \frac{1}{k} > 0 \Rightarrow k > 0$
- $-\frac{1}{k} > -\frac{1}{2} \Rightarrow \frac{1}{k} < \frac{1}{2} \Rightarrow k > 2$

$\Rightarrow$  Unstable open loop is stabilized when in closed loop configuration for  $k \in [0, 2]$ .

Why this works: Cauchy's Argument principle

•  $G(s)$  "maps" points:



• The mapping is characterized by poles & zeros

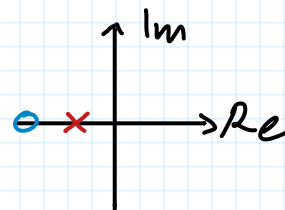
$$\Rightarrow |G(s)| = \frac{\sum' |s - z_i|}{\sum' |s - p_i|}$$

$$\Rightarrow \angle G(s) = \sum' \angle(s - z_i) - \sum' \angle(s - p_i)$$

So for the mapping of a point  $s$ , we must evaluate its phasors with respect to each pole / zero!

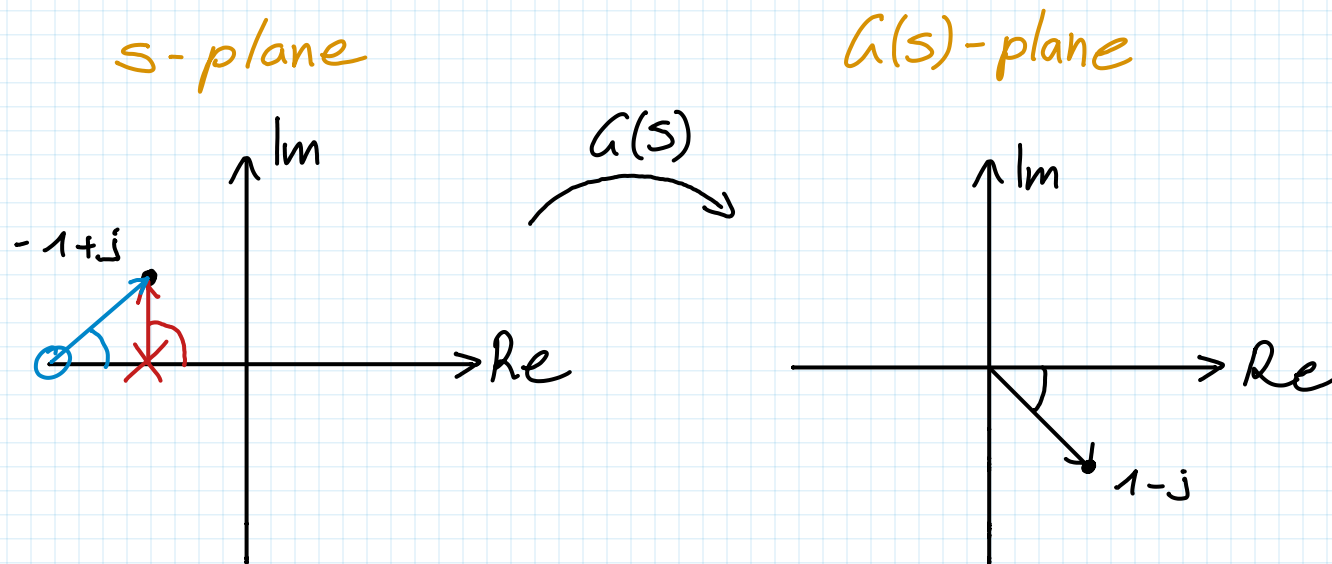
Example:

$$G(s) = \frac{s+2}{s+1}$$



$$\bullet |G(s)| = \frac{|s+2|}{|s+1|} \quad , \quad \bullet \angle G(s) = \angle(s+2) - \angle(s+1)$$

Now let's see how points are mapped:



$$|G(-1+j)| = \frac{|-1+j+2|}{|-1+j+1|} = \frac{\sqrt{1+1}}{1} = \sqrt{2}$$

$$\begin{aligned} \angle G(-1+j) &= \angle(-1+j+2) - \angle(-1+j+1) \\ &= \arctan(1) - \arctan(\infty) = 45^\circ - 90^\circ = -45^\circ \end{aligned}$$

Core takeaway, zeros "add" phase, poles "subtract" it.

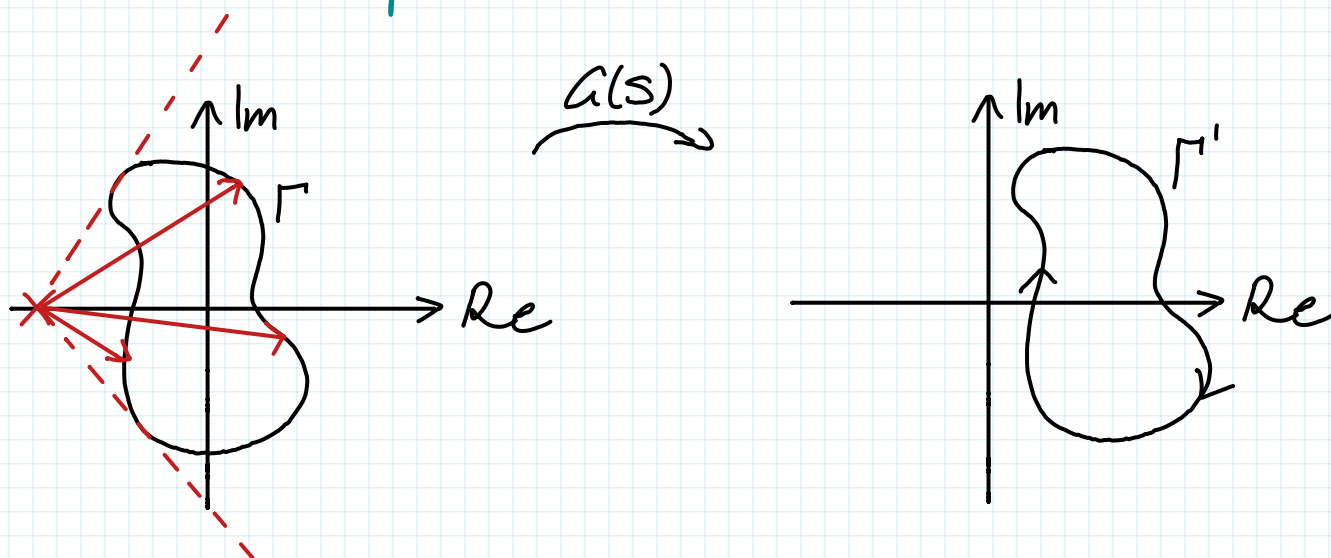
- We can now do this for many points. Especially interesting results will arise when we do this for infinitesimally close points, along a closed, directional curve.

$\Rightarrow$  Contour  $\Gamma$

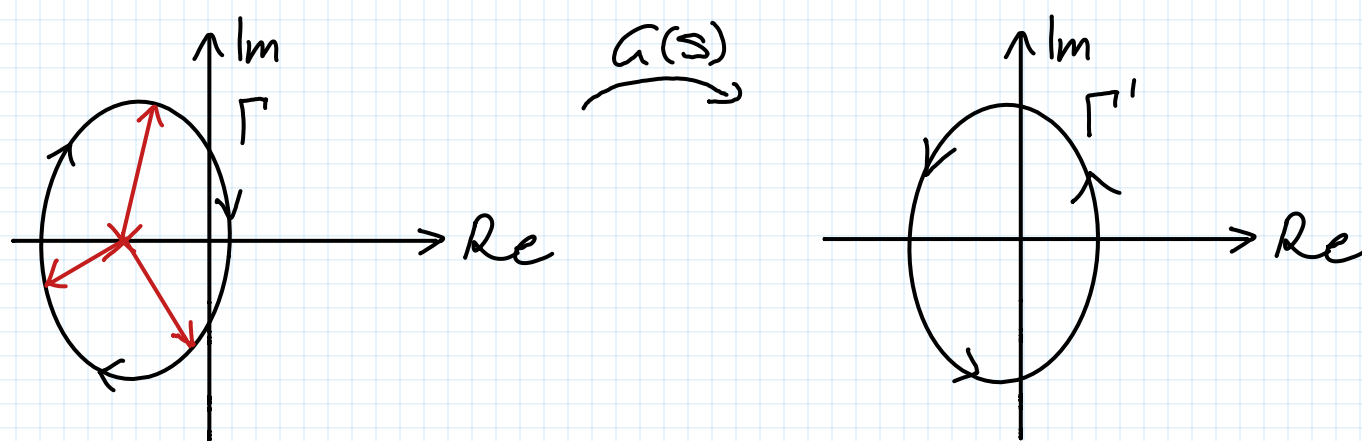
Let's start with a simpler system with a single pole  $p$ .

$$G(s) = \frac{1}{s-p}$$

First, let's draw a contour  $\Gamma$  that doesn't encircle the pole.

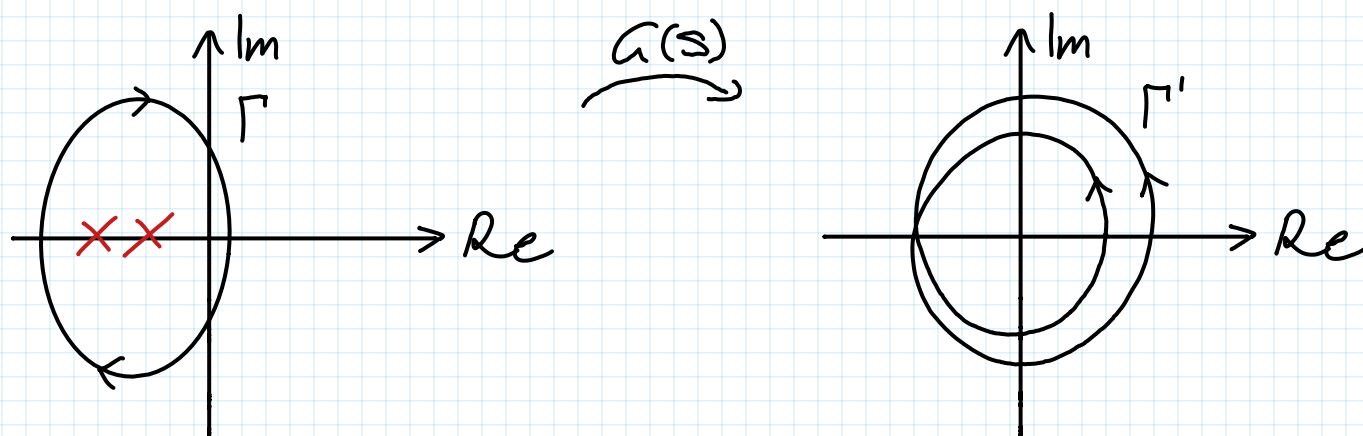


• If we choose the contour  $\Gamma$  to encircle the pole...



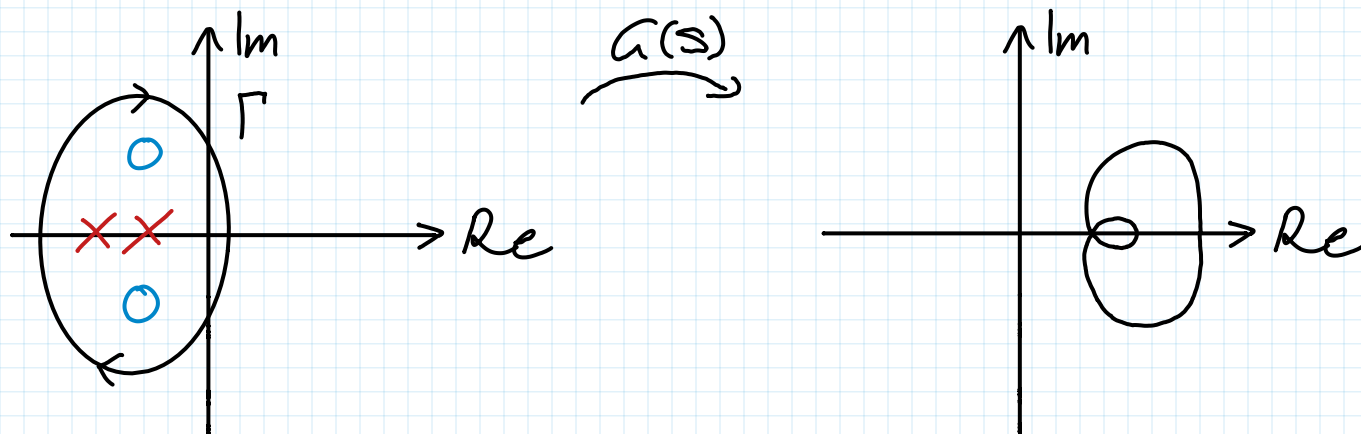
$\Gamma'$  will encircle the origin in the opposite direction!

• What if we add two poles?



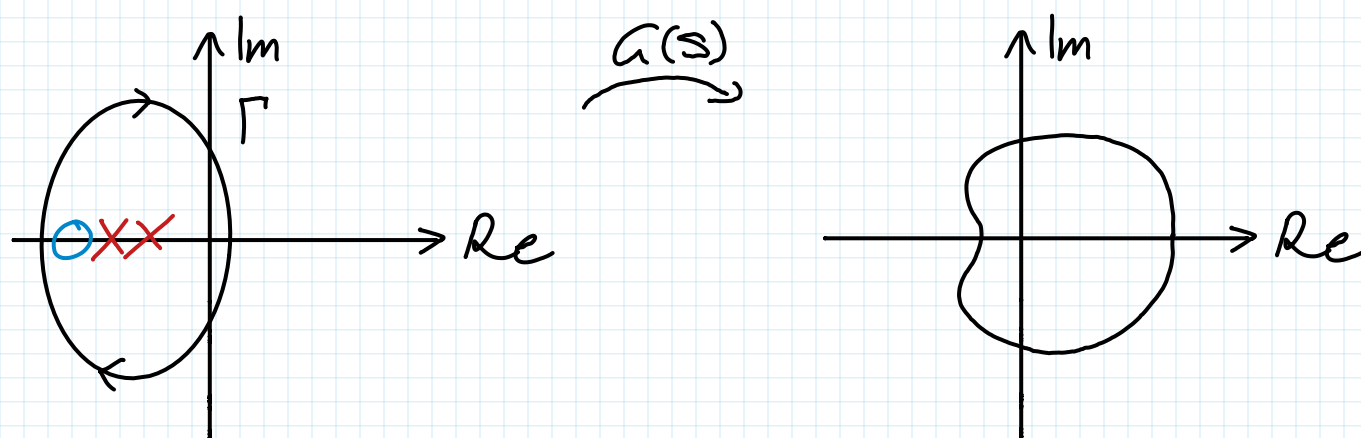
$\Rightarrow$  Two encirclements of the origin!

• And what if there are zeros?



Here the zeros add  $+2 \cdot 360^\circ$  and the poles subtract  $-2 \cdot 360^\circ$ . Therefore there are no origin encirclements.

• So what if we have an unequal amount of  $p_i$  and  $z_i$ ?



We conclude:

$$N = Z - P$$

$N$ : Number of times  $G(s)$  encircles origin  
 $Z$ : Number of zeros of  $G(s)$   
 $P$ : Number of poles of  $G(s)$

Okay, this is neat - but how does it help us design controllers?

## From Cauchy's Argument principle to Nyquist's stability theorem

- Recall the closed-loop tf

$$T(s) = \frac{kL(s)}{1 + kL(s)}$$

Note: Here for simplicity  
 $L(s) \hat{=}$  Plant, dynamic part  
 of controller

Unstable if  $\text{poles}(T(s)) > 0$

But poles of  $T(s)$  are really the zeros of  $1 + kL(s)$ .

|                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------|
| $\begin{aligned} \# \text{RHP zeros of } 1 + kL(s) \\ \hat{=} \\ \# \text{unstable closed-loop poles} \end{aligned}$ |
|----------------------------------------------------------------------------------------------------------------------|

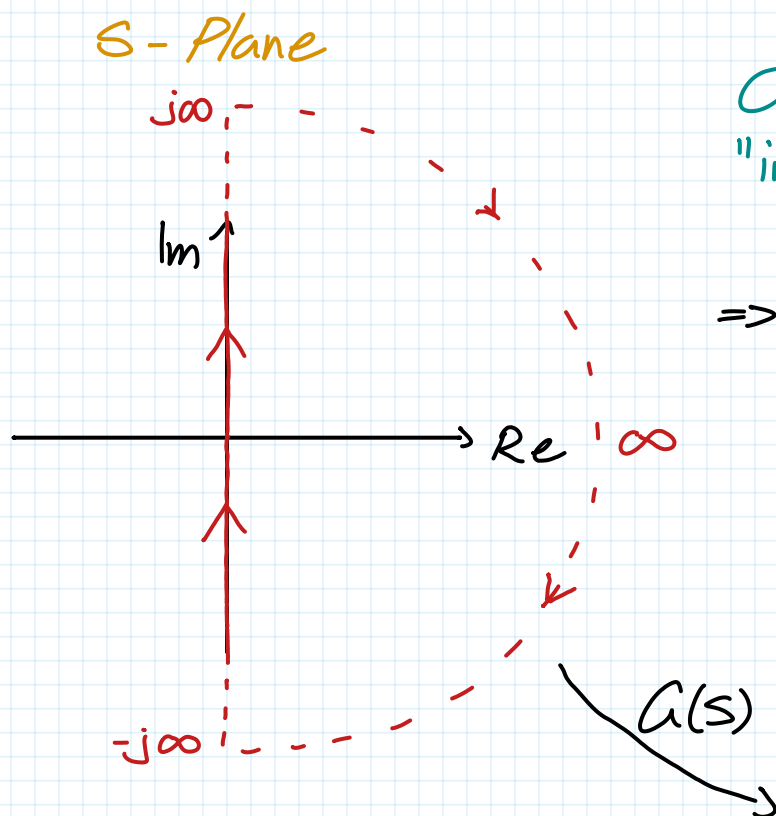
Core idea:

Use Cauchy's Argument principle, to check for RHP zeros of  $1 + L(s)$ :

$\Rightarrow$  Draw a contour in the complex plane, that entails information about the total RHP...



=> Solution: Choose the Nyquist-contour or D-contour:

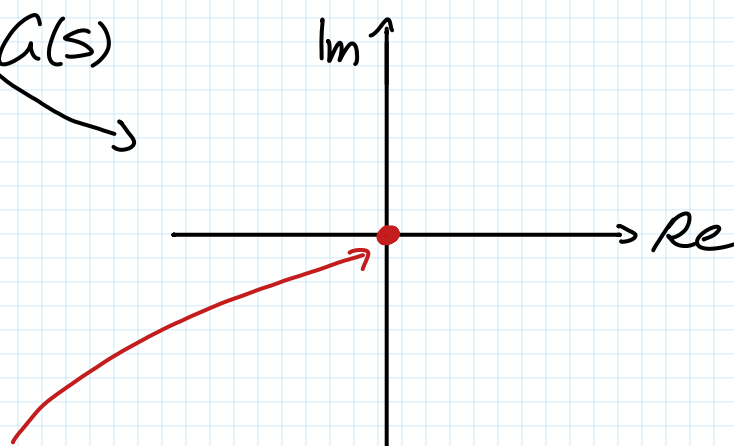


Close the contour at "infinity"

=> Now choose the mapping as

$$G(s) = 1 + kL(s)$$

**G(s)-plane**



From  $N = Z - P$ :

# Encirclements of origin of  $1 + kL(s)$  plot (I)

(II)

$\triangleq$

(III)

# RHP zeros of  $1 + kL(s)$  - RHP poles of  $1 + kL(s)$

• This can be reformulated to become more practical. First, notice that looking for solutions of

$$1 + kL(s) = 0 \quad \Leftrightarrow \quad L(s) = -\frac{1}{k}$$

↑  
origin

Therefore we can rewrite

Simplifies (I)

$$\begin{aligned} &\# \text{Encirclements of origin of } 1+kL(s) \text{ plot} \\ &\quad \hat{=} \\ &\# \text{Encirclements of } -\frac{1}{k} \text{ point of } L(s) \text{ plot} \end{aligned}$$

• Secondly, recall that we can write

$$L(s) = \frac{N(s)}{\boxed{D(s)}}$$

And therefore

$$1+kL(s) = 1+k \frac{N(s)}{D(s)} = \frac{D(s)+kN(s)}{\boxed{D(s)}}$$

$\Rightarrow$  Denominator (poles) of  $L(s)$  are unaffected by closing the loop

Therefore we can write:

$$\begin{aligned} &\# \text{RHP poles of } 1+kL(s) \\ &\quad \hat{=} \\ &\# \text{Unstable open-loop poles} \end{aligned}$$

Simplifies (II)

• Finally, from the closed-loop tf

$$T(s) = \frac{kL(s)}{1+kL(s)}$$

We have that zeros of  $1 + kL(s)$  correspond to the poles of  $T(s) = \frac{kL(s)}{1 + kL(s)}$ .

$$\begin{array}{c} \# \text{RHP zeros of } 1 + kL(s) \\ \hat{=} \\ \# \text{Unstable closed loop poles} \end{array}$$

Simplifies (III)

Let us combine everything to a final statement:

Nyquist stability criterion

$$Z = N - P$$

$N$ : # of encirclements of  $-\frac{1}{k}$

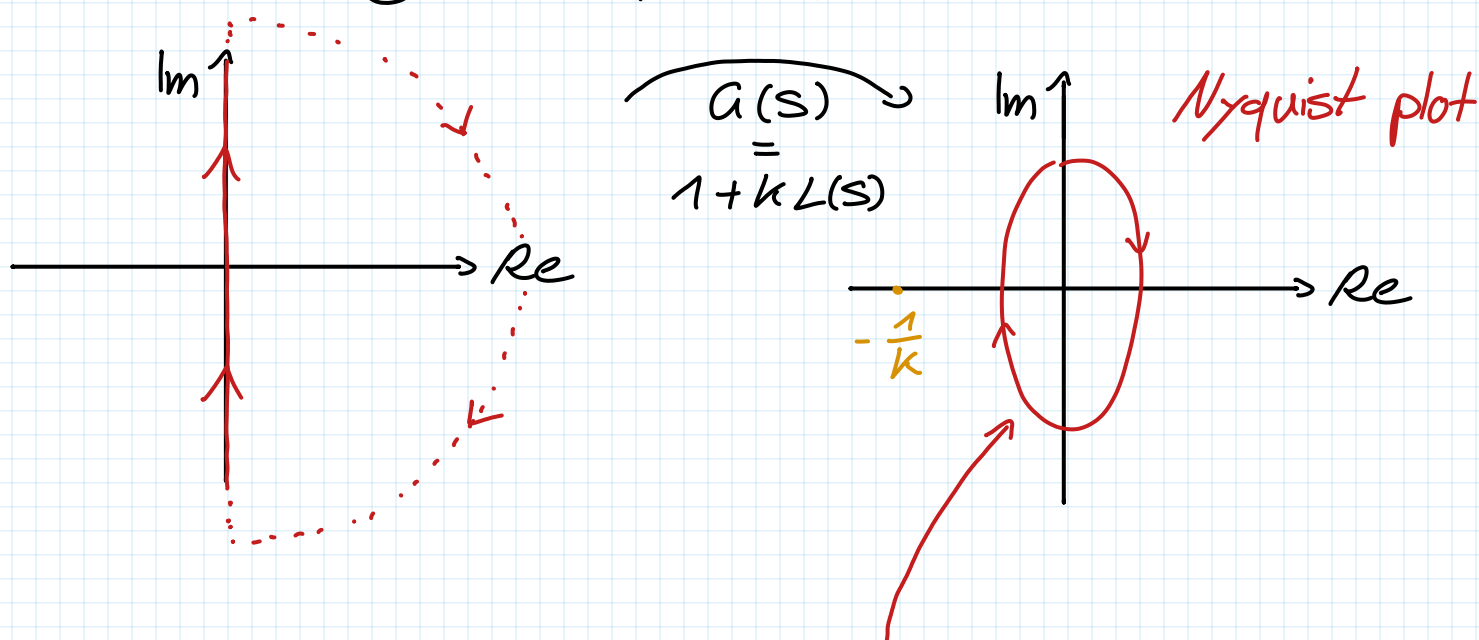
$P$ : # of unstable open-loop poles

$Z$ : # of unstable closed-loop poles

$\Rightarrow$  **Unknown** closed-loop stability can be assessed using the **known** open-loop system!

That was a lot. Let's summarize the important points.

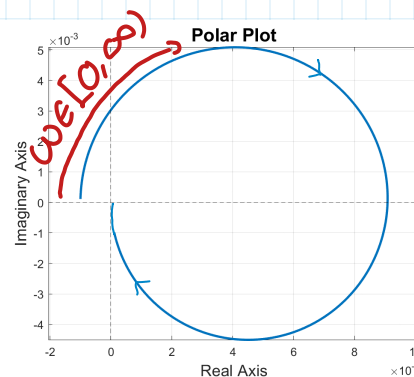
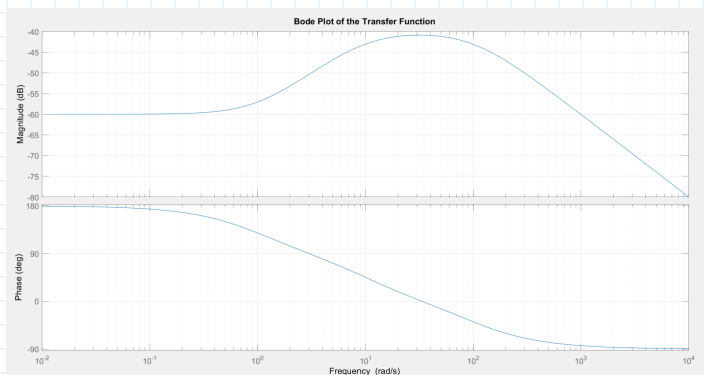
- When choosing to map a Nyquist-contour via  $G(s)$



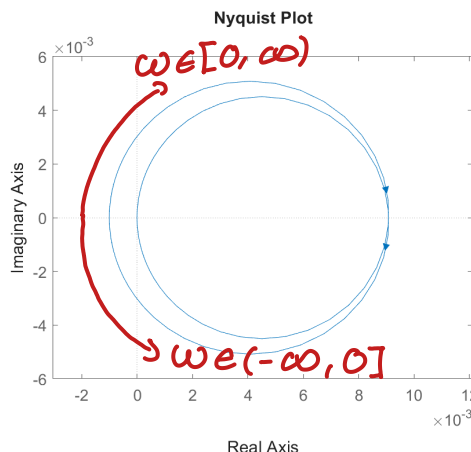
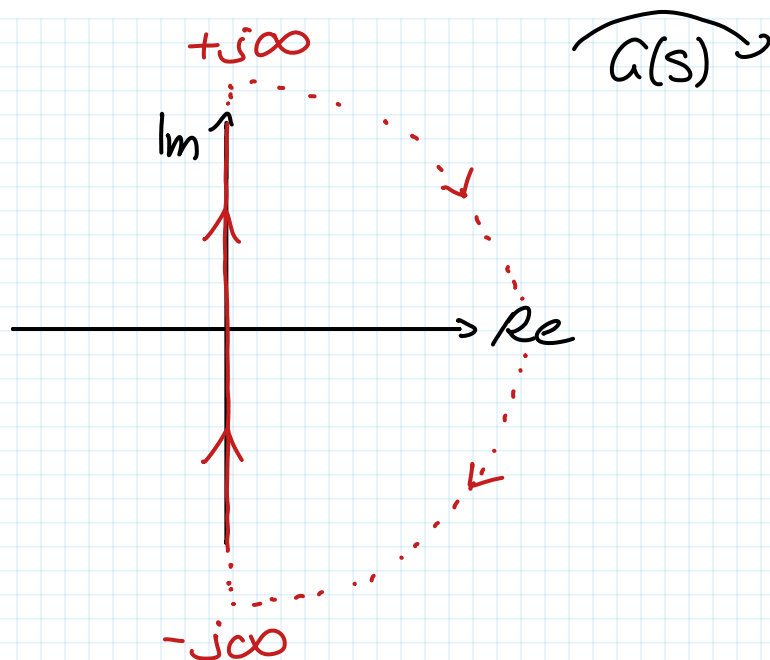
we get a shape called the Nyquist plot, that depends on the poles and zeros of  $G(s)$ .

- Now we can assess stability of the closed-loop, by evaluating the Nyquist criterion  $N = Z - P$ .

Recall: Earlier we said that the Nyquist plot is just an extension of the polar plot, which itself is just a reformulation of the Bode plot...



$\omega \in [0, \infty)$



⇒ Mirroring the polar plot to get the Nyquist plot is exactly the same as mapping the Nyquist-contour!

### In essence

- We want to assess the closed-loop, based on the open-loop frequency response.

⇒ Bode plot works often, but not always!

- So we introduced the Nyquist plot.
- Using the Nyquist stability criterion on the Nyquist plot, closed loop stability can always be assessed!

### Why not just use root locus?

- + Nyquist also works when only experimental data present.
- + Root locus can't give insights on robustness.
- + Nyquist can even handle non-rational transfer functions.

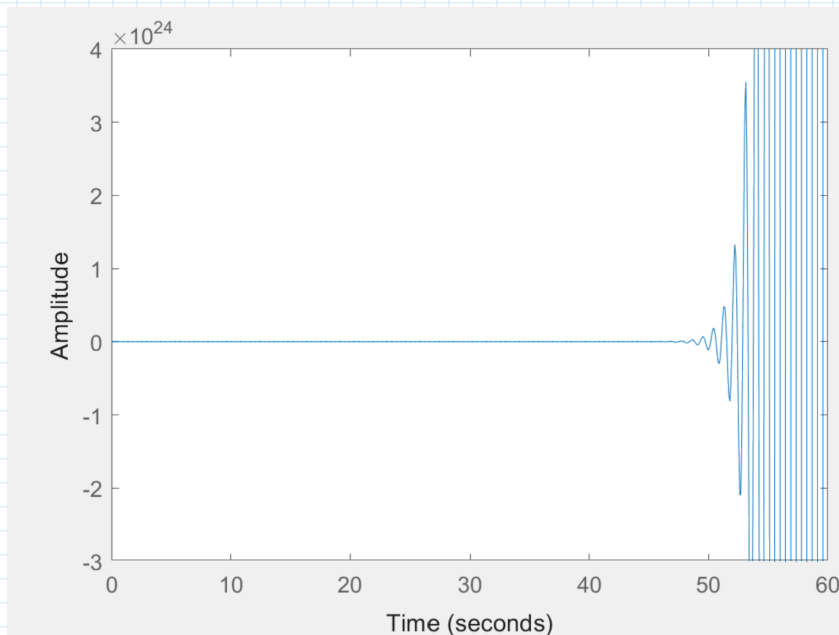
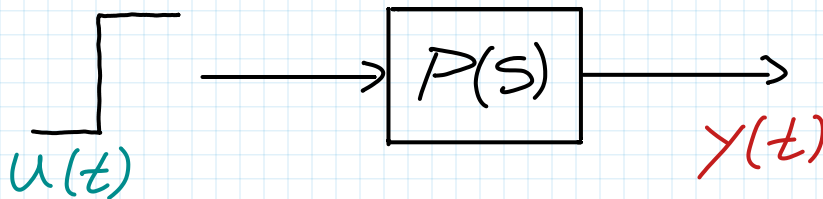
Another example:

$$\text{Given } P(s) = \frac{s^2 + 2s - 3}{s^3 - 2s^2 + 50s + 10}$$

zeros:  $z_1 = -3, z_2 = 1$

poles:  $p_1 \approx 1 + 7i, p_2 \approx 1 - 7i, p_3 \approx -0,2$

- Unstable poles, non-minimumphase zeros  
 $\Rightarrow$  overall undesirable behaviour



Step response  
 $y(t)$  of  $P(s)$

• Can we stabilize this?

$\Rightarrow$  Apply Nyquist:  $N = Z - P$

For closed-loop stability we need:

$$Z = 0$$

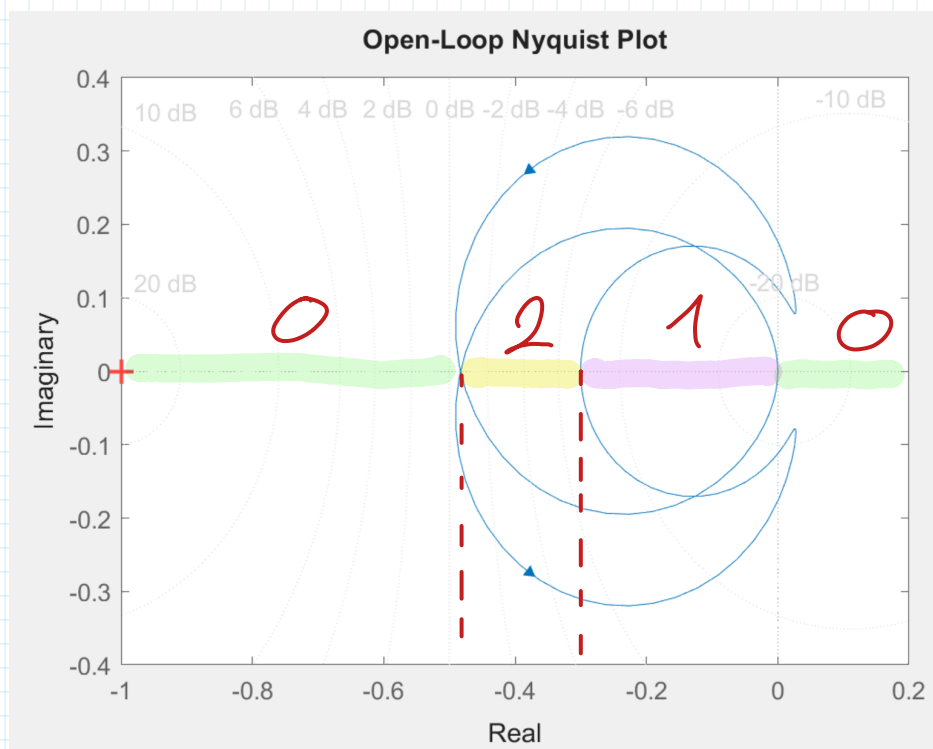
We have 2 unstable poles:

$$P = 2$$

Therefore we need:

$$N = -2 \quad (2, \text{ counter-clockwise})$$

• Now we examine the Nyquist plot of  $P(s)$ :



$\Rightarrow$  To stabilize,  $-1/k$  must be within this area

From the plot we approximate

$$\cdot -\frac{1}{k} < -0,3 \Rightarrow k \gtrsim 3.3$$

$$\cdot -\frac{1}{k} > -0,47 \Rightarrow k \lesssim 2.1$$

Matlab

$\Rightarrow$  System is stabilized with a simple scalar gain  $k \in [2.1, 3.3]$  in a standard feedback configuration

With a perfectly known system and no disturbances this works great.

In reality though, we must not only ensure stability, but also robustness.



# Robustness

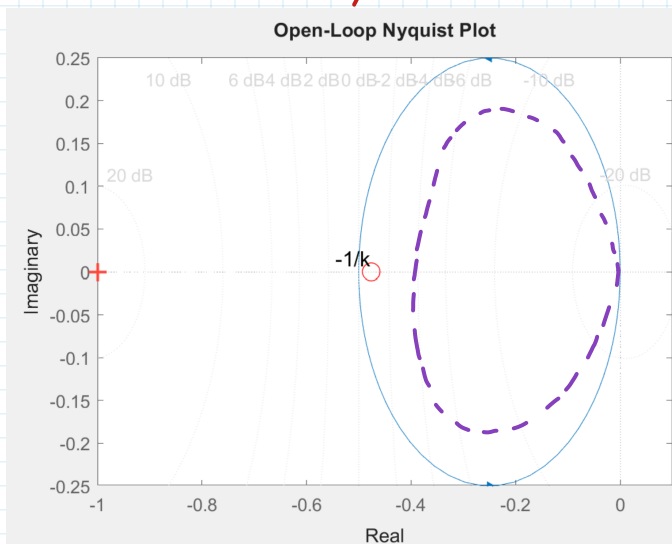
This term refers to "how" stable our system is. A system is robust, if large disturbances and uncertainties don't make it unstable.

- Robustness can be measured by how close the Nyquist plot comes to the  $-1/k$  point.

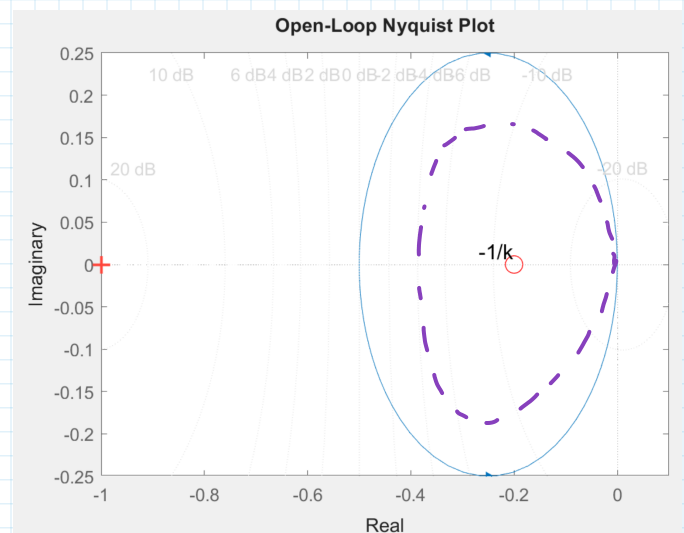
Ex:  $P(s) = \frac{1}{s-2}$

$k > 2$

$k = 2.05$



$k = 5$

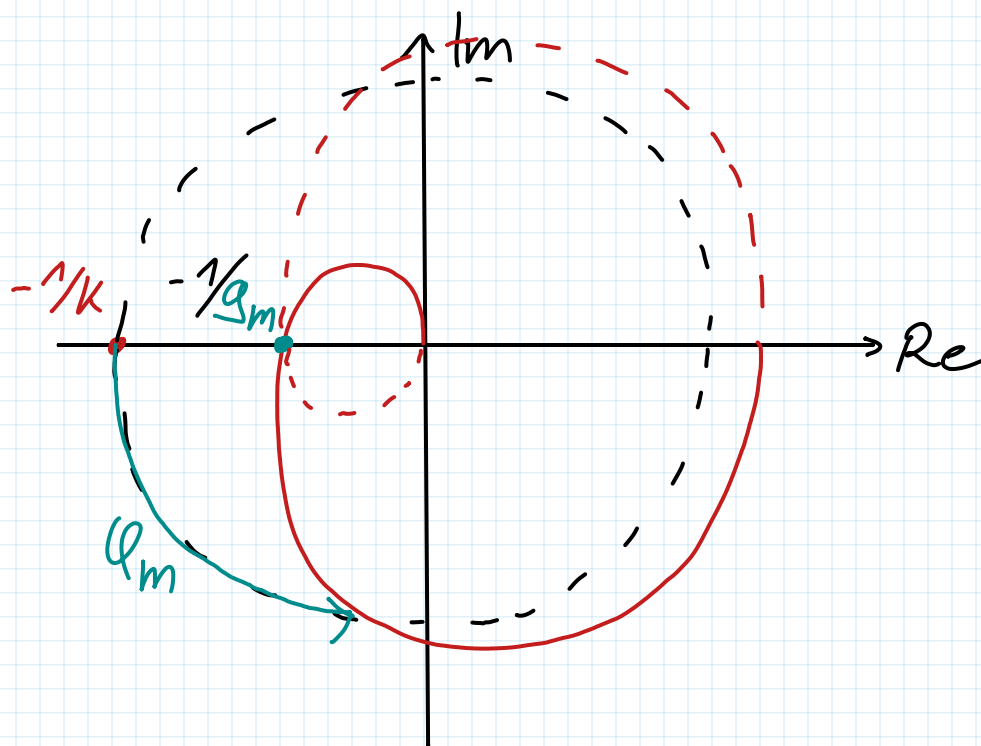


Disturbed system becomes unstable for  $k = 2.05$ , but not for  $k = 5$

• This "degree of stability" can be measured in the form of so called stability margins.

Let's visualize these margins with an example:

Assume an open-loop stable system  
 $\Rightarrow P = 0$ , hence to remain stable  
 $(Z = 0)$ , we need  $N = 0$ .



If the plot encircles  $-1/k$  the closed loop becomes unstable!

$\Rightarrow$  We measure the "safety distance" to  $-1/k$  with:

$g_m$ : Gain margin,  $q_m$ : Phase margin

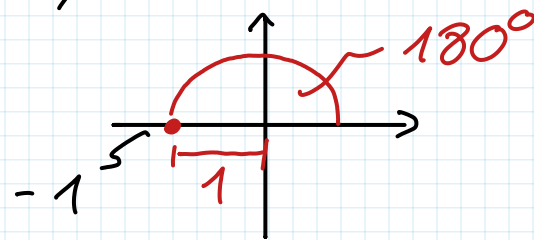
We want to keep the margins high!

- Gain and phase margins can also be observed in the open-loop Bode plot.

⇒ The system operates at the unstable  $-1$  point, if:

- $|L(s)| = 1$

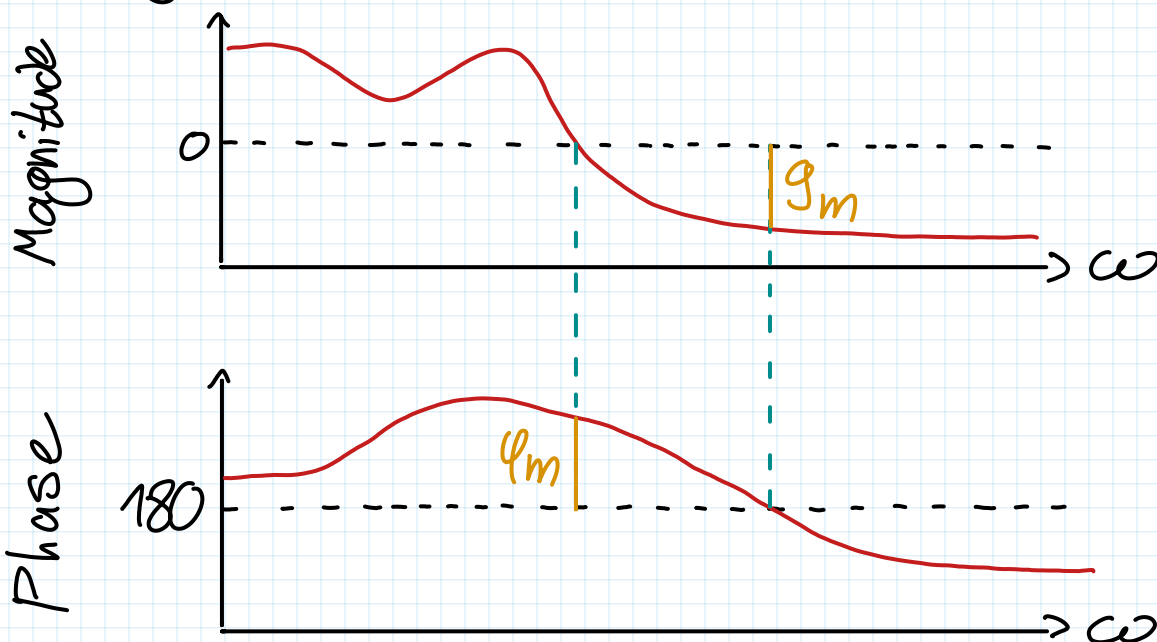
- $\angle L(s) = 180^\circ$



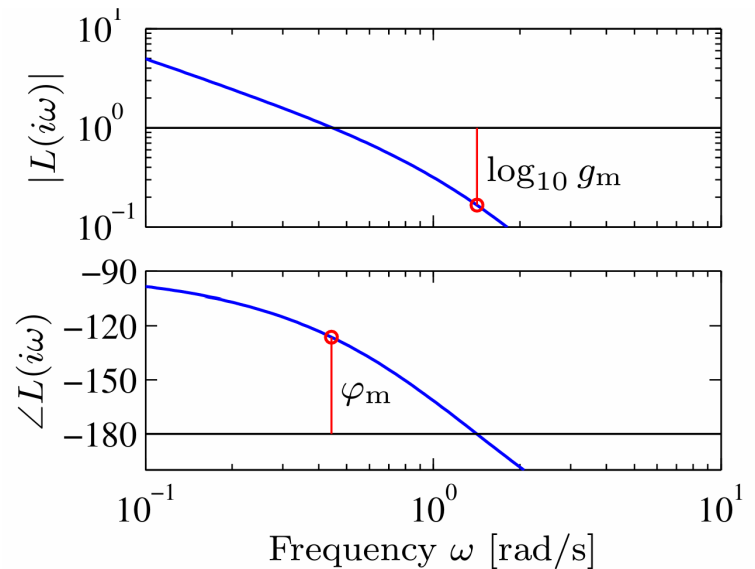
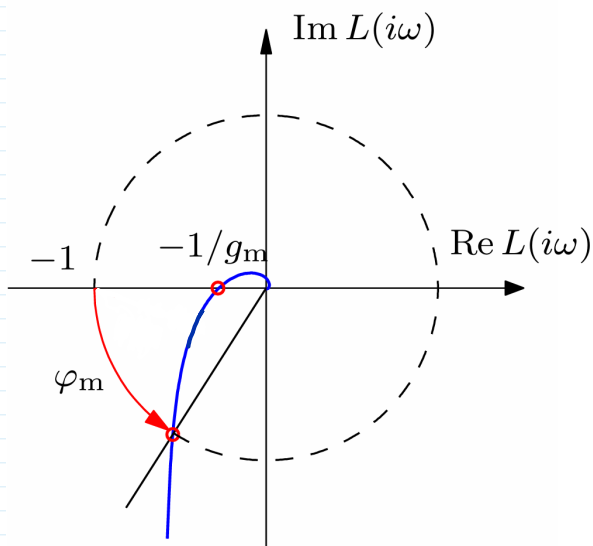
- So to ensure stability, we need

$$|L(s)| < 1, \text{ whenever } \angle L(s) = 180^\circ$$

- On the Bode plot, we see the margins as such:



Bringing it all together:



Now you know all the major plots in CSI:

- Root locus
- Bode
- Nyquist

Together they hold great power to analyze systems.

Me  
explaining  
Nyquist

